Quantum Information Processing (2021) 20:282
https://doi.org/10.1007/s11128-021-03216-8

Study of decoherence and memory in modified Eisert–Wilkens–Lewenstein scheme

A. V. S. Kameshwari[1] · S. Balakrishnan[1]

Received: 7 December 2020 / Accepted: 11 August 2021 / Published online: 1 September 2021
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2021

Abstract
Eisert–Wilkens–Lewenstein scheme is the well-known quantization scheme that exploits only controlled unitary operators due to the requirement of having a classical game as a subset of the quantum version. By revoking this requirement, the modified Eisert–Wilkens–Lewenstein scheme is proposed to admit the other two-qubit entangling operators. In the present work, we introduce the notions of decoherence and memory in the modified Eisert–Wilkens–Lewenstein scheme. While decoherence is known to decrease the payoffs of the players, memory increases it. Further, the present work reveals the availability of suitable entangling operators, which can increase (decrease) the effect of memory (decoherence) on the average payoff of the players under the given game setting.

Keywords Quantum games · Modified EWL scheme · Entangling operators · Decoherence · Memory

1 Introduction

Classical game theory is a study of mathematical models of the strategic interaction between the group members [1]. It helps us to understand human behavior when they try to make decisions in conflict situations. The year 1944 marks a significant development of game theory with the publication of "The Theory of Games and Economic Behavior" by John von Neumann and Oskar Morgenstern [2]. This development made game theory an essential tool for research in various disciplines like economics, political sciences, psychology and medicine [3]. Recent studies have shown the application

✉ S. Balakrishnan
physicsbalki@gmail.com

A. V. S. Kameshwari
kameshwariavs0@gmail.com

1 Department of Physics, School of Advanced Sciences, Vellore Institute of Technology, Vellore, Tamil Nadu 632014, India

of quantum mechanics in classical game theory [3]. The notion of quantum game theory is introduced by David Meyer [4]. Quantum game theory has got much attention from the work of Meyer and by the introduction of the quantization scheme [5]. With the help of the quantization scheme, one can achieve a superior Nash Equilibrium in the quantum games [6]. Classically, the presence of communication gap between the players brings a dilemma in the game. It is resolved for the two parametric strategic space, as studied by Eisert et al. [5, 7], and for one parametric strategic space [8, 9]. Entanglement, a weird characteristic of microscopic particles exhibited by quantum mechanics, is used very extensively to show the prowess of quantum game theory [4–11]. Upon tuning the degree of entanglement, one can appreciate the advantage of a quantum player over a classical player [7, 10]. Refer [11] for the experimental realization of quantum games and the usefulness of entanglement in the quantum games.

A classical game is extended to the domain of quantum mechanics with the help of a quantization scheme. Eisert–Wilkens–Lewenstein scheme (hereafter, EWL) [5] and Marinatto Weber scheme (MW) [12] are the two available quantization schemes. Both the schemes are used extensively to understand the quantum version of the classical game. It is to be mentioned that the EWL scheme considers classical game as a subset of the quantum game due to which only one class of entangling operators, namely controlled unitaries are considered in the quantization.

It is to be noted that when quantum mechanics is introduced to the classical game, the quantum strategy defeats the classical strategy [13]. Entanglement is found to bring the desired outcome in terms of payoff to the player who employs it. Such an understanding is shown to be invalid in the recent work on the quantum two-penny flip problem [14]. It is shown in [14] that entanglement, although superior, failed to help the winning of the quantum player, and eventually, the two-penny flip problem faces a classical game-like situation in the quantum game setup. Also, in the work of Eisert and Wilkens [15], it is shown that quantum entanglement alone cannot resolve the prisoner's dilemma.

Decoherence, also known as noise, causes loss of information when the system interacts with the environment. The influence of decoherence comes into the picture when the quantum information is transferred through the communication channel interacting with the environment. Therefore, noise is an unwanted disturbance in the communication process. In the context of game theory, decoherence reduces the benefits (payoffs) of the interacting agents, refer [16–20] for instance. Memory is considered as one of the parameters which play a very crucial role in evaluating the payoff of the players. To be precise, memory is the player's knowledge of past successful choices used to optimize their respective strategies [20–23]. The noisy memoryless channels are called uncorrelated, and the noisy channels with memory are called correlated. Detailed study of Nawaz reveals that memory controls the payoff reduction due to decoherence and for the limit of maximum memory, decoherence becomes ineffective [22].

Modified EWL scheme is introduced by revoking the requirement of having a classical game as a subset of the quantum game; refer to [24] for detailed analysis. The modified EWL scheme admits a wide range of entangling operators in the quantum setting of classical games. Such an attempt would help to find the usefulness of the entangling operators in the research work on game theory. In the present work, we

study the symmetric games, in particular prisoner's dilemma, using modified EWL scheme. The motive behind this study is to understand the role of the arbiter, who employs the entangling operator, in changing the dynamics of the game. Further, we introduce the notions of decoherence and memory in the modified EWL scheme to understand their influence on the payoffs of the players. We have brought out some interesting observations in this line of thought, and the results are discussed in detail.

2 Modified EWL scheme

Historically the well-known EWL quantization scheme has faced a few criticisms, one of them being, the limitation to understand the quantumness of the game when a classical game is a subset of a quantum game [25]. To appreciate the quantum nature of the game, the existing EWL scheme is modified by relaxing the commutator relations. In this scheme, the game starts with the arbiter preparing the initially entangled state using a two-qubit non-local entangling operator. This state is then passed to the players, namely Alice and Bob, who perform their strategies. Generally, players can adapt any of the strategies, namely σ_x and I, where I corresponds to the classical strategy cooperation (C), and σ_x corresponds to defection (D) in the prisoner's dilemma game. The players can also resort to strategies $C = I$, $D = i\sigma_y$ and $Q = i\sigma_z$ where Q is Alice's best response to Bob's Strategy D according to the EWL scheme [5]. After the applications of the strategies by the players, a disentangling operator is applied. The general form of the two-qubit operator is:

$$J(c_1, c_2, c_3) = \begin{pmatrix} e^{-\frac{c_3}{2}}c^- & 0 & 0 & -ie^{-\frac{c_3}{2}}s^- \\ 0 & e^{\frac{c_3}{2}}c^+ & -ie^{\frac{c_3}{2}}s^+ & 0 \\ 0 & -ie^{\frac{c_3}{2}}s^+ & e^{\frac{c_3}{2}}c^+ & 0 \\ -ie^{-\frac{c_3}{2}}s^- & 0 & 0 & e^{-\frac{c_3}{2}}c^- \end{pmatrix} \tag{1}$$

where $c^{\pm} = \cos\left(\frac{c_1 \pm c_2}{2}\right)$;$s^{\pm} = \sin\left(\frac{c_1 \pm c_2}{2}\right)$ and c_1, c_2, c_3 are the geometrical points of the two-qubit non-local operator J satisfying the condition $c_1 \geq c_2 \geq c_3 \geq 0$ [26, 27]. In the case of EWL scheme, the requirement of having a classical game as a subset of the quantum game is imposed by the computation conditions between the entangling operators J and the pure strategies, namely σ_x and I to be zero. Due to this, entangling operator J reduces to controlled unitary operators, $J(c_1, 0, 0)$, where $0 \leq c_1 \leq \pi/2$ and note that $J(\pi/2, 0, 0)$ corresponds to Controlled-NOT operation. It is to be mentioned that we have considered $c_3 = 0$ in Eq. (1) for the sake of mathematical simplicity in all the calculations. In other words, two-qubit entangling operators is given by $J(c_1, c_2, 0)$.

The general quantum strategic space of Alice and Bob are the single-qubit local operator belonging to the SU (2) group, it is given in [24]

$$U_A(\theta_1, \alpha_1, \beta_1) = \begin{pmatrix} e^{i\alpha_1}\cos\left(\frac{\theta_1}{2}\right) & ie^{i\beta_1}\sin\left(\frac{\theta_1}{2}\right) \\ ie^{-i\beta_1}\sin\left(\frac{\theta_1}{2}\right) & e^{-i\alpha_1}\cos\left(\frac{\theta_1}{2}\right) \end{pmatrix};$$

$$U_B(\theta_2, \alpha_2, \beta_2) = \begin{pmatrix} e^{i\alpha_2}\cos\left(\frac{\theta_2}{2}\right) & ie^{i\beta_2}\sin\left(\frac{\theta_2}{2}\right) \\ ie^{-i\beta_2}\sin\left(\frac{\theta_2}{2}\right) & e^{-i\alpha_2}\cos\left(\frac{\theta_2}{2}\right) \end{pmatrix} \tag{2}$$

where $0 \leq \theta_i \leq \pi$, $-\pi \leq \alpha_i \leq \pi$, and $-\pi \leq \beta_i \leq \pi$ with $i = 1, 2$. The tensor product of single-qubit operators generates the two-qubit local operator. It is known that only a two-qubit non-local operator produces entanglement between the qubits. The initial state of the game $|\psi_i\rangle$ is given by:

$$|\psi_i\rangle = a_1a_2|00\rangle + a_1b_2|01\rangle + b_1a_2|10\rangle + b_1b_2|11\rangle \tag{3}$$

Here the coefficients satisfy the normalization condition, i.e., $|a_1a_2|^2 + |a_1b_2|^2 + |b_1a_2|^2 + |b_1b_2|^2 = 1$.

3 Modified EWL scheme with Decoherence

When the physical system interacts with the environment, decoherence occurs [16]. Essentially, decoherence or noise affects the communication process. In the game-theoretic structure, decoherence reduces the payoff of the players [17]. Since decoherence is an inevitable process, it is necessary to find a way to reduce its effect on the game dynamics. In this section, the average payoff of the players is analyzed for the noisy communication channel using the modified EWL scheme. In order to see this, the condition for the equivalence of the average payoff of the players is calculated and is shown for the case when amplitude damping is the noisy channel, and both Alice and Bob are quantum players. The available Kraus operators for the amplitude damping channel are [16],

$$m_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix} \text{ and } m_1 = \begin{pmatrix} 0 & \sqrt{p} \\ 0 & 0 \end{pmatrix}$$

where p is the measure of decoherence, i.e., noise in the channel and it ranges from zero to one. While $p = 0$ refers to a noiseless channel, $p = 1$ indicates a maximal noisy channel. The Kraus operator m_0 is used in this work as we can realize the effect of decoherence for all the initial states, but this is not the case with the operator m_1. Unlike m_0, the Kraus operator m_1, due to its matrix form does not show any significant change in the average payoffs for all the initial states.

3.1 Amplitude damping in channel 1

In this case, the condition for the equivalence of the average payoff of the players is analyzed, when decoherence is in only channel 1 as shown in Fig. 1.

The final state of the game for this case is represented as:

$$\left|\psi_f\right\rangle = J^\dagger(U_A \otimes U_B)(m_0 \otimes m_0)J|\psi_i\rangle \tag{4}$$

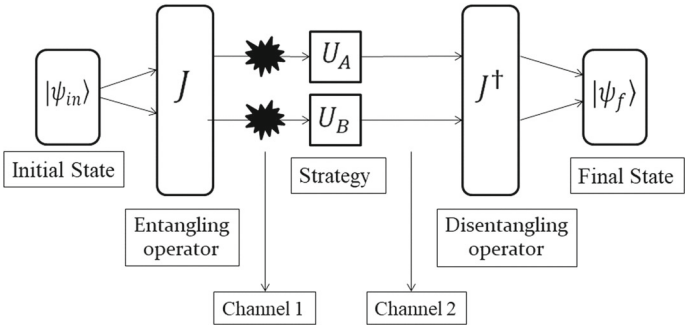

Fig. 1 Modified EWL scheme with amplitude damping in channel 1

where U_A and U_B are strategies of Alice and Bob as given in Eq. (2), and m_0 is the Kraus operator for amplitude damping. Upon substituting quantum strategies U_A and U_B and the Kraus operator m_0, $(U_A \otimes U_B)(m_0 \otimes m_0)$ becomes:

$$\begin{pmatrix} e^{i(\alpha_1+\alpha_2)}c_1c_2 & i\sqrt{1-p}e^{i(\alpha_1+\beta_2)}c_1s_2 & i\sqrt{1-p}e^{i(\alpha_2+\beta_1)}s_1c_2 & -(1-p)e^{i(\beta_1+\beta_2)}s_1s_2 \\ ie^{i(\alpha_1-\beta_2)}c_1s_2 & \sqrt{1-p}e^{i(\alpha_1-\alpha_2)}c_1c_2 & -\sqrt{1-p}e^{i(\beta_1-\beta_2)}s_1s_2 & i(1-p)e^{i(\beta_1-\alpha_2)}s_1c_2 \\ ie^{i(\alpha_2-\beta_1)}s_1c_2 & -\sqrt{1-p}e^{i(\beta_2-\beta_1)}s_1s_2 & \sqrt{1-p}e^{i(\alpha_2-\alpha_1)}c_1c_2 & i(1-p)e^{i(\beta_2-\alpha_1)}c_1s_2 \\ -e^{-i(\beta_1+\beta_2)}s_1s_2 & i\sqrt{1-p}e^{-i(\alpha_2+\beta_1)}s_1c_2 & i\sqrt{1-p}e^{-i(\alpha_1+\beta_2)}c_1s_2 & (1-p)e^{-i(\alpha_1+\alpha_2)}c_1c_2 \end{pmatrix} \tag{5}$$

where $c_1 = \cos\left(\frac{\theta_1}{2}\right)$; $c_2 = \cos\left(\frac{\theta_2}{2}\right)$; $s_1 = \sin\left(\frac{\theta_1}{2}\right)$ and $s_2 = \sin\left(\frac{\theta_2}{2}\right)$ are the elements of the two-qubit local operator. The above expression after simplification is written as:

$$(U_A \otimes U_B)(m_0 \otimes m_0) = \begin{pmatrix} A & i\sqrt{1-p}E & i\sqrt{1-p}H^* & -(1-p)D^* \\ iB & \sqrt{1-p}F & -\sqrt{1-p}G^* & i(1-p)C^* \\ iC & -\sqrt{1-p}G & \sqrt{1-p}F^* & i(1-p)B^* \\ -D & i\sqrt{1-p}H & i\sqrt{1-p}E^* & (1-p)A^* \end{pmatrix} \tag{6}$$

The final state of the game for this case is obtained by substituting two-qubit entangling operator from Eq. (1) with $c_3 = 0$, initial state from Eqs. (3) and (6) in Eq. (4). After calculating the final state in the computational basis $|00\rangle$, $|01\rangle$, $|10\rangle$ and $|11\rangle$, we get:

$$\begin{aligned} \langle\psi_f|00\rangle = {} & a_1a_2\left(A^*c^-c^- + iD^*s^-c^- - i(1-p)Dc^-s^- + (1-p)As^-s^-\right) \\ & + a_1b_2\left(-i\sqrt{1-p}E^*c^-c^+ - \sqrt{1-p}H^*s^-c^+ + \sqrt{1-p}Hc^-s^+ - i\sqrt{1-p}Es^-s^+\right) \\ & + b_1a_2\left(\sqrt{1-p}E^*c^-s^+ - i\sqrt{1-p}H^*s^-s^+ - i\sqrt{1-p}Hc^-c^+ - \sqrt{1-p}Es^-c^+\right) \\ & + b_1b_2\left(iA^*c^-s^- - D^*s^-s^- - (1-p)Dc^-c^- - i(1-p)As^-c^-\right) \end{aligned} \tag{7}$$

$$\langle\psi_f|01\rangle = a_1a_2\left(-iB^*c^+c^- - C^*s^+c^- + (1-p)Cc^+s^- - i(1-p)Bs^+s^-\right)$$

$$+ a_1 b_2\left(\sqrt{1-p}F^* c^+ c^+ + i\sqrt{1-p}G^* s^+ c^+ - i\sqrt{1-p}Gc^+ s^+ + \sqrt{1-p}Fs^+ s^+\right)$$
$$+ b_1 a_2\left(i\sqrt{1-p}F^* c^+ s^+ - \sqrt{1-p}G^* s^+ s^+ - \sqrt{1-p}Gc^+ c^+ - i\sqrt{1-p}Fs^+ c^+\right)$$
$$+ b_1 b_2\left(B^* c^+ s^- - iC^* s^+ s^- - i(1-p)Cc^+ c^- - (1-p)Bs^+ c^-\right) \tag{8}$$

$$\begin{aligned}\langle\psi_f|10\rangle = {} & a_1 a_2\left(-B^* s^+ c^- - iC^* c^+ c^- - i(1-p)Cs^+ s^- + (1-p)Bc^+ s^-\right)\\ & + a_1 b_2\left(-i\sqrt{1-p}F^* s^+ c^+ - \sqrt{1-p}G^* c^+ c^+ - \sqrt{1-p}Gs^+ s^+ + i\sqrt{1-p}Fc^+ s^+\right)\\ & + b_1 a_2\left(\sqrt{1-p}F^* s^+ s^+ - i\sqrt{1-p}G^* c^+ s^+ + i\sqrt{1-p}Gs^+ c^+ + \sqrt{1-p}Fc^+ c^+\right)\\ & + b_1 b_2\left(-iB^* s^+ s^- + C^* c^+ s^- - (1-p)Cs^+ c^- - i(1-p)Bc^+ c^-\right)\end{aligned} \tag{9}$$

$$\begin{aligned}\langle\psi_f|11\rangle = {} & a_1 a_2\left(-iA^* s^- c^- - D^* c^- c^- - (1-p)Ds^- s^- + i(1-p)Ac^- s^-\right)\\ & + a_1 b_2\left(-\sqrt{1-p}E^* s^- c^+ - i\sqrt{1-p}H^* c^- c^+ - i\sqrt{1-p}Hs^- s^+ + \sqrt{1-p}Ec^- s^+\right)\\ & + b_1 a_2\left(-i\sqrt{1-p}E^* s^- s^+ + \sqrt{1-p}H^* c^- s^+ - \sqrt{1-p}Hs^- c^+ - i\sqrt{1-p}Ec^- c^+\right)\\ & + b_1 b_2\left(A^* s^- s^- - iD^* c^- s^- + i(1-p)Ds^- c^- + (1-p)Ac^- c^-\right)\end{aligned} \tag{10}$$

From the above expression, the average payoff of Alice and Bob is calculated for a symmetric game where X, Y, W and Z are the elements of the payoff matrix. Note that a symmetric game is a game in which the same strategy is available to all the players. For example, the prisoner's dilemma game and the chicken game are symmetric games. Symmetric game is represented by the following payoff matrix:

$$P_{\text{Alice}} = \begin{pmatrix} X & Y \\ W & Z \end{pmatrix} \; P_{\text{Bob}} = \begin{pmatrix} X & W \\ Y & Z \end{pmatrix}$$

The result is compiled in Table 1 for all the initial states and the strategies I and σ_x adapted by the players. It is worth mentioning that the result is the same when players adapt the strategies $C = I$, $D = i\sigma_y$ and $Q = i\sigma_z$ as like original Eisert–Wilkens–Lewenstein scheme. This is due to the fact that the expression of average payoff contains modulus term which kills the extra phase factor present in the strategies D and Q.

In all the analysis, we consider the prisoner's dilemma problem using the payoff matrix:

$$P_{\text{Alice}} = \begin{pmatrix} 3 & 0 \\ 5 & 1 \end{pmatrix}$$

From Table 1, it is observed that for the initial state $|00\rangle$ and $|11\rangle$, the average payoff of the players is equal i.e., $\$_A = \$_B$ for the entangling operator $J\left(c_1 = c_2 = \frac{\pi}{4}, c_3 = 0\right)$ independent of the strategies adapted by the players. The identified entangling operator $J\left(\frac{\pi}{4}, \frac{\pi}{4}, 0\right)$ is recognized as a Double CNOT. Further, for the initial state $|01\rangle$ and $|10\rangle$, when the players adapt different strategies, it is observed that the expression for the average payoff of the players is equal for any choice of entangling operator $J(c_1, c_2 = c_1, c_3 = 0)$. When the players adapt the

Table 1 Average payoff of Alice and Bob in case of Amplitude damping in channel 1

Initial state	Strategies	Average payoff of Alice $\$_A$	Average payoff of Bob $\$_B$
$\|00\rangle$	$I \otimes I$	$X\left(c^-c^- + (1-p)s^-s^-\right)^2 + Z\left(s^-c^- - (1-p)c^-s^-\right)^2$	$X\left(c^-c^- + (1-p)s^-s^-\right)^2 + Z\left(s^-c^- - (1-p)c^-s^-\right)^2$
	$I \otimes \sigma_x$	$Y\left(c^+c^- + (1-p)s^+s^-\right)^2 + W\left(s^+c^- - (1-p)c^+s^-\right)^2$	$W\left(c^+c^- + (1-p)s^+s^-\right)^2 + Y\left(s^+c^- - (1-p)c^+s^-\right)^2$
	$\sigma_x \otimes I$	$Y\left(s^+c^- - (1-p)c^+s^-\right)^2 + W\left(c^+c^- + (1-p)s^+s^-\right)^2$	$W\left(s^+c^- - (1-p)c^+s^-\right)^2 + Y\left(c^+c^- + (1-p)s^+s^-\right)^2$
	$\sigma_x \otimes \sigma_x$	$X\left(c^-s^- - (1-p)c^-s^-\right)^2 + Z\left(c^-c^- + (1-p)s^-s^-\right)^2$	$X\left(c^-s^- - (1-p)c^-s^-\right)^2 + Z\left(c^-c^- + (1-p)s^-s^-\right)^2$
$\|11\rangle$	$I \otimes I$	$X\left(c^-s^- - (1-p)c^-s^-\right)^2 + Z\left(s^-s^- + (1-p)c^-c^-\right)^2$	$X\left(c^-s^- - (1-p)c^-s^-\right)^2 + Z\left(s^-s^- + (1-p)c^-c^-\right)^2$
	$I \otimes \sigma_x$	$Y\left(s^-c^+ - (1-p)s^+c^-\right)^2 + W\left(s^+s^- + (1-p)c^+c^-\right)^2$	$W\left(s^-c^+ - (1-p)s^+c^-\right)^2 + Y\left(s^+s^- + (1-p)c^+c^-\right)^2$
	$\sigma_x \otimes I$	$Y\left(s^+s^- + (1-p)c^+c^-\right)^2 + W\left(c^+s^- - (1-p)s^+c^-\right)^2$	$W\left(s^+s^- + (1-p)c^+c^-\right)^2 + Y\left(c^+s^- - (1-p)s^+c^-\right)^2$
	$\sigma_x \otimes \sigma_x$	$X\left(s^-s^- + (1-p)c^-c^-\right)^2 + Z\left(c^-s^- - (1-p)c^-s^-\right)^2$	$X\left(s^-s^- + (1-p)c^-c^-\right)^2 + Z\left(c^-s^- - (1-p)c^-s^-\right)^2$
$\|01\rangle$	$I \otimes I$	$Y(1-p)$	$W(1-p)$
	$I \otimes \sigma_x$	$X\left[(1-p)\left(c^-c^+ + s^-s^+\right)^2\right] + Z\left[(1-p)\left(s^-c^+ - c^-s^+\right)^2\right]$	$X\left[(1-p)\left(c^-c^+ + s^-s^+\right)^2\right] + Z\left[(1-p)\left(s^-c^+ - c^-s^+\right)^2\right]$
	$\sigma_x \otimes I$	$X\left[(1-p)\left(s^-c^+ - c^-s^+\right)^2\right] + Z\left[(1-p)\left(c^-c^+ + s^-s^+\right)^2\right]$	$X\left[(1-p)\left(s^-c^+ - c^-s^+\right)^2\right] + Z\left[(1-p)\left(c^-c^+ + s^-s^+\right)^2\right]$
	$\sigma_x \otimes \sigma_x$	$W(1-p)$	$Y(1-p)$
$\|10\rangle$	$I \otimes I$	$W(1-p)$	$Y(1-p)$
	$I \otimes \sigma_x$	$X\left[(1-p)\left(c^-s^+ - s^-c^+\right)^2\right] + Z\left[(1-p)\left(s^-s^+ + c^-c^+\right)^2\right]$	$X\left[(1-p)\left(c^-s^+ - s^-c^+\right)^2\right] + Z\left[(1-p)\left(s^-s^+ + c^-c^+\right)^2\right]$
	$\sigma_x \otimes I$	$X\left[(1-p)\left(s^-s^+ + c^-c^+\right)^2\right] + Z\left[(1-p)\left(c^-s^+ - s^-c^+\right)^2\right]$	$X\left[(1-p)\left(s^-s^+ + c^-c^+\right)^2\right] + Z\left[(1-p)\left(c^-s^+ - s^-c^+\right)^2\right]$
	$\sigma_x \otimes \sigma_x$	$Y(1-p)$	$W(1-p)$

same strategies (either $I \otimes I$ or $\sigma_x \otimes \sigma_x$), the expression for the average payoff of the players is independent of the entangling operator. Such an observation can be found in the recents works of Elgazzar [8, 9]. Thus, role of the arbiter who applies the entangling operator is restricted to the initial states $|01\rangle$ and $|10\rangle$. Thus, the significance of the entangling operators depends on the initial state of the game.

Using the expressions of the average payoffs of Alice, the prisoner's dilemma game is analyzed for the entangling operator J and decoherence parametrized by p for the different strategies and various initial states. In order to overcome the mathematical difficulties, we have considered $c_2 = \frac{\pi}{4}$, and hence, the two-qubit operators assume the form $J\left(\frac{\pi}{2} \geq c_1 \geq \frac{\pi}{4}, c_2 = \frac{\pi}{4}, c_3 = 0\right)$ in all the analysis. It is evident from the all the plots that when there is no noise in the channel ($p = 0$), the average payoff of Alice remains unaffected. It is also observed from Fig. 2b–d that for a given value of c_1 (for a particular entangling operator), the average payoff decreases with the increase in decoherence. This is true for all the strategies of the initial states $|01\rangle, |10\rangle, |11\rangle$. Interestingly, for the initial state $|00\rangle$, when the strategies are $\sigma_x \otimes \sigma_x$, increase in

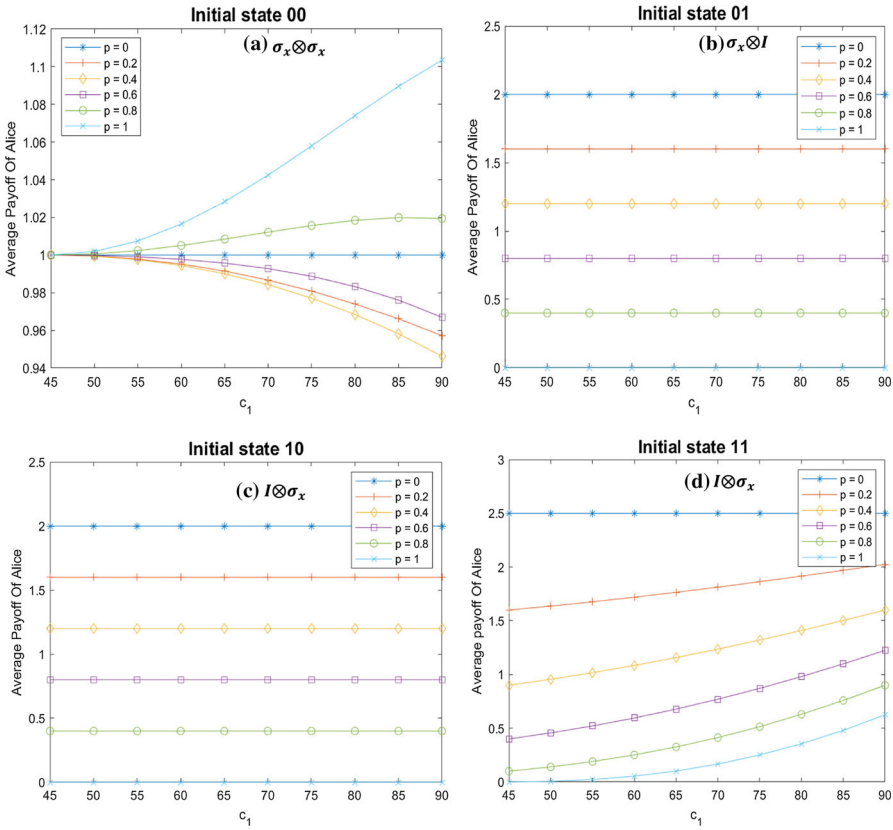

Fig. 2 Average payoff of Alice for different initial states and strategies when amplitude damping is in channel 1

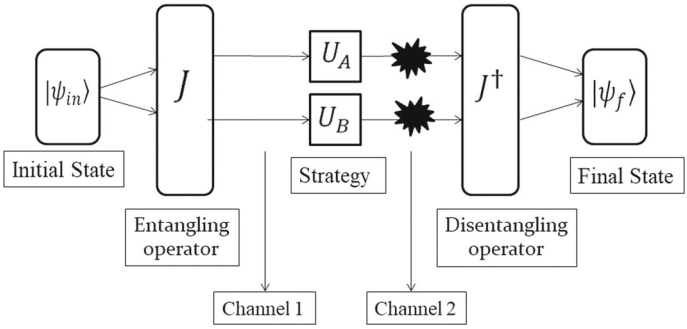

Fig. 3 Modified EWL scheme with amplitude damping in channel 2

decoherence can increase the average payoff. We can realize this fact by observing Fig. 2a closely that when p is greater (smaller) than 0.6, then the average payoff is more (less) than what it can be at $p = 0$. This unexpected behavior of average payoff with increasing decoherence for the initial state $|00\rangle$ is due to the dependence of the average payoff on the strategy, initial state and the entangling operator J. Similar observations can be found in the work of Flitney and Abbott [17] on quantum prisonner's dilemma problem, where Bob plays quantum strategies, and Alice is restricted to classical strategies. The increase in payoff with the increase in decoherence is also observed by Salman et al., for the maximally entangled initial state due to constructive interference of quantum phase for the amplitude damping channel [18].

3.2 Amplitude damping in channel 2

In this case, the condition for the equivalence of the average payoff of the player is analyzed when decoherence is in channel 2. Figure 3 shows the modified EWL scheme with the amplitude damping in channel 2.

The final state of the game under this condition is represented as

$$\left|\psi_f\right\rangle = J^\dagger (m_0 \otimes m_0)(U_A \otimes U_B) J |\psi_i\rangle \tag{11}$$

where $(m_0 \otimes m_0)(U_A \otimes U_B)$ becomes

$$\begin{pmatrix} e^{i(\alpha_1+\alpha_2)}c_1c_2 & ie^{i(\alpha_1+\beta_2)}c_1s_2 & ie^{i(\alpha_2+\beta_1)}s_1c_2 & -e^{i(\beta_1+\beta_2)}s_1s_2 \\ i\sqrt{1-p}e^{i(\alpha_1-\beta_2)}c_1s_2 & \sqrt{1-p}e^{i(\alpha_1-\alpha_2)}c_1c_2 & -\sqrt{1-p}e^{i(\beta_1-\beta_2)}s_1s_2 & i\sqrt{1-p}e^{i(\beta_1-\alpha_2)}s_1c_2 \\ i\sqrt{1-p}e^{i(\alpha_2-\beta_1)}s_1c_2 & -\sqrt{1-p}e^{i(\beta_2-\beta_1)}s_1s_2 & \sqrt{1-p}e^{i(\alpha_2-\alpha_1)}c_1c_2 & i\sqrt{1-p}e^{i(\beta_2-\alpha_1)}c_1s_2 \\ -(1-p)e^{i(\beta_1+\beta_2)}s_1s_2 & i(1-p)e^{-i(\alpha_2+\beta_1)}s_1c_2 & i(1-p)e^{-i(\alpha_1+\beta_2)}c_1s_2 & (1-p)e^{-i(\alpha_1+\alpha_2)}c_1c_2 \end{pmatrix} \tag{12}$$

The above expression after simplification is written as:

$$(U_A \otimes U_B)(m_0 \otimes m_0) = \begin{pmatrix} A & iE & iH^* & -D^* \\ i\sqrt{1-p}B & \sqrt{1-p}F & -\sqrt{1-p}G^* & i\sqrt{1-p}C^* \\ i\sqrt{1-p}C & -\sqrt{1-p}G & \sqrt{1-p}F^* & i\sqrt{1-p}B^* \\ -(1-p)D & i(1-p)H & i(1-p)E^* & (1-p)A^* \end{pmatrix} \tag{13}$$

The final state of the game for this case is obtained by substituting two-qubit entangling operator from Eq. (1) with $c_3 = 0$, initial state from Eqs. (3) and (13) in Eq. (11). After calculating the final state in the computational basis $|00\rangle$, $|01\rangle$, $|10\rangle$ and $|11\rangle$, we get:

$$\begin{aligned} \langle\psi_f|00\rangle = {} & a_1a_2\big(A^*c^-c^- + i(1-p)D^*s^-c^- - iDc^-s^- + (1-p)As^-s^-\big) \\ & + a_1b_2\big(-iE^*c^-c^+ - (1-p)H^*s^-c^+ + Hc^-s^+ - i(1-p)Es^-s^+\big) \\ & + b_1a_2\big(E^*c^-s^+ - i(1-p)H^*s^-s^+ - iHc^-c^+ - (1-p)Es^-c^+\big) \\ & + b_1b_2\big(iA^*c^-s^- - (1-p)D^*s^-s^- - Dc^-c^- - i(1-p)As^-c^-\big) \end{aligned} \tag{14}$$

$$\begin{aligned} \langle\psi_f|01\rangle = {} & a_1a_2\Big(-i\sqrt{1-p}B^*c^+c^- - \sqrt{1-p}C^*s^+c^- + \sqrt{1-p}Cc^+s^- - i\sqrt{1-p}Bs^+s^-\Big) \\ & + a_1b_2\Big(\sqrt{1-p}F^*c^+c^+ + i\sqrt{1-p}G^*s^+c^+ - i\sqrt{1-p}Gc^+s^+ + \sqrt{1-p}Fs^+s^+\Big) \\ & + b_1a_2\Big(i\sqrt{1-p}F^*c^+s^+ - \sqrt{1-p}G^*s^+s^+ - \sqrt{1-p}Gc^+c^+ - i\sqrt{1-p}Fs^+c^+\Big) \\ & + b_1b_2\Big(\sqrt{1-p}B^*c^+s^- - i\sqrt{1-p}C^*s^+s^- - i\sqrt{1-p}Cc^+c^- - \sqrt{1-p}Bs^+c^-\Big) \end{aligned} \tag{15}$$

$$\begin{aligned} \langle\psi_f|10\rangle = {} & a_1a_2\Big(-\sqrt{1-p}B^*s^+c^- - i\sqrt{1-p}C^*c^+c^- - i\sqrt{1-p}Cs^+s^- + \sqrt{1-p}Bc^+s^-\Big) \\ & + a_1b_2\Big(-i\sqrt{1-p}F^*s^+c^+ - \sqrt{1-p}G^*c^+c^+ - \sqrt{1-p}Gs^+s^+ + i\sqrt{1-p}Fc^+s^+\Big) \\ & + b_1a_2\Big(\sqrt{1-p}F^*s^+s^+ - i\sqrt{1-p}G^*c^+s^+ + i\sqrt{1-p}Gs^+c^+ + \sqrt{1-p}Fc^+c^+\Big) \\ & + b_1b_2\Big(-i\sqrt{1-p}B^*s^+s^- + \sqrt{1-p}C^*c^+s^- - \sqrt{1-p}Cs^+c^- - i\sqrt{1-p}Bc^+c^-\Big) \end{aligned} \tag{16}$$

$$\begin{aligned} \langle\psi_f|11\rangle = {} & a_1a_2\big(-iA^*s^-c^- - (1-p)D^*c^-c^- - Ds^-s^- + i(1-p)Ac^-s^-\big) \\ & + a_1b_2\big(-E^*s^-c^+ - i(1-p)H^*c^-c^+ - iHs^-s^+ + (1-p)Ec^-s^+\big) \\ & + b_1a_2\big(-iE^*s^-s^+ + (1-p)H^*c^-s^+ - Hs^-c^+ - i(1-p)Ec^-c^+\big) \\ & + b_1b_2\big(A^*s^-s^- - i(1-p)D^*c^-s^- + iDs^-c^- + (1-p)Ac^-c^-\big) \end{aligned} \tag{17}$$

From the above expressions, the average payoff of Alice and Bob is calculated for the prisoner's dilemma game where X, Y, W and Z are the elements of the payoff matrix. The result is compiled in Table 2 for all the initial states and different strategies when amplitude damping is in channel 2.

From Table 2, it is observed that for the initial state $|00\rangle$ and $|11\rangle$, the average payoff of the players is equal i.e., $\$_A = \$_B$ for the entangling operator $J\big(\frac{\pi}{2} \geq c_1 \geq \frac{\pi}{4}, c_2 = \frac{\pi}{4}, c_3 = 0\big)$ independent of the strategies adapted by the players.

Table 2 Average payoff of Alice and Bob in case of Amplitude damping in channel 2

Initial state	Strategies	Average payoff of Alice $\$_A$	Average payoff of Bob $\$_B$
$\|00\rangle$	$I \otimes I$	$X\left(c^-c^- + (1-p)s^-s^-\right)^2 + Z\left(s^-c^- - (1-p)c^-s^-\right)^2$	$X\left(c^-c^- + (1-p)s^-s^-\right)^2 + Z\left(s^-c^- - (1-p)c^-s^-\right)^2$
	$I \otimes \sigma_x$	$Y\left[(1-p)\left(c^-c^+ + s^+s^-\right)^2\right] + W\left[(1-p)\left(s^+c^- - c^+s^-\right)^2\right]$	$W\left[(1-p)\left(c^-c^+ + s^+s^-\right)^2\right] + Y\left[(1-p)\left(s^+c^- - c^+s^-\right)^2\right]$
	$\sigma_x \otimes I$	$Y\left[(1-p)\left(s^+c^- - c^+s^-\right)^2\right] + W\left[(1-p)\left(c^+c^- + s^+s^-\right)^2\right]$	$W\left[(1-p)\left(s^+c^- - c^+s^-\right)^2\right] + Y\left[(1-p)\left(c^+c^- + s^+s^-\right)^2\right]$
	$\sigma_x \otimes \sigma_x$	$X\left(c^-s^- - (1-p)s^-c^-\right)^2 + Z\left(s^-s^- + (1-p)c^-c^-\right)^2$	$X\left(c^-s^- - (1-p)s^-c^-\right)^2 + Z\left(s^-s^- + (1-p)c^-c^-\right)^2$
$\|11\rangle$	$I \otimes I$	$X\left(c^-s^- - (1-p)s^-c^-\right)^2 + Z\left(s^-s^- + (1-p)c^-c^-\right)^2$	$X\left(c^-s^- - (1-p)s^-c^-\right)^2 + Z\left(s^-s^- + (1-p)c^-c^-\right)^2$
	$I \otimes \sigma_x$	$Y\left[(1-p)\left(c^+s^- - s^+c^-\right)^2\right] + W\left[(1-p)\left(s^+s^- + c^+c^-\right)^2\right]$	$W\left[(1-p)\left(c^+s^- - s^+c^-\right)^2\right] + Y\left[(1-p)\left(s^+s^- + c^+c^-\right)^2\right]$
	$\sigma_x \otimes I$	$Y\left[(1-p)\left(c^+c^- + s^+s^-\right)^2\right] + W\left[(1-p)\left(c^+s^- - s^+c^-\right)^2\right]$	$W\left[(1-p)\left(c^+c^- + s^+s^-\right)^2\right] + Y\left[(1-p)\left(c^+s^- - s^+c^-\right)^2\right]$
	$\sigma_x \otimes \sigma_x$	$X\left(c^-c^- + (1-p)s^-s^-\right)^2 + Z\left(s^-c^- - (1-p)c^-s^-\right)^2$	$X\left(c^-c^- + (1-p)s^-s^-\right)^2 + Z\left(s^-c^- - (1-p)c^-s^-\right)^2$
$\|01\rangle$	$I \otimes I$	$Y(1-p)$	$W(1-p)$
	$I \otimes \sigma_x$	$X\left(c^-c^+ + (1-p)s^-s^+\right)^2 + Z\left(s^-c^+ - (1-p)c^-s^+\right)^2$	$X\left(c^-c^+ + (1-p)s^-s^+\right)^2 + Z\left(s^-c^+ - (1-p)c^-s^+\right)^2$
	$\sigma_x \otimes I$	$X\left(c^-s^+ - (1-p)s^-c^+\right)^2 + Z\left(s^-s^+ + (1-p)c^-c^+\right)^2$	$X\left(c^-s^+ - (1-p)s^-c^+\right)^2 + Z\left(s^-s^+ + (1-p)c^-c^+\right)^2$
	$\sigma_x \otimes \sigma_x$	$W(1-p)$	$Y(1-p)$
$\|10\rangle$	$I \otimes I$	$W(1-p)$	$Y(1-p)$
	$I \otimes \sigma_x$	$X\left(c^-s^+ - (1-p)s^-c^+\right)^2 + Z\left(s^-s^+ + (1-p)c^-c^+\right)^2$	$X\left(c^-s^+ - (1-p)s^-c^+\right)^2 + Z\left(s^-s^+ + (1-p)c^-c^+\right)^2$
	$\sigma_x \otimes I$	$X\left(c^-c^+ + (1-p)s^-s^+\right)^2 + Z\left(s^-c^+ - (1-p)c^-s^+\right)^2$	$X\left(c^-c^+ + (1-p)s^-s^+\right)^2 + Z\left(s^-c^+ - (1-p)c^-s^+\right)^2$
	$\sigma_x \otimes \sigma_x$	$Y(1-p)$	$W(1-p)$

The identified entangling operator belongs to the class of entangling operators known as perfect entanglers. Perfect entanglers are the quantum gates that generate maximally entangled states for some initially separable states [27]. Further, for the initial states $|01\rangle$ and $|10\rangle$, arbiter who applies the entangling operator becomes insignificant when the players adapt the same strategies (either $I \otimes I$ or $\sigma_x \otimes \sigma_x$). The role of arbiter when amplitude damping is in channel 2 that is similar to the case when amplitude damping is in channel 1.

Using the expressions of the average payoffs of Alice shown in Table 2, the prisoner's dilemma game is analyzed for entangling operator J and decoherence parametrized by p for the different strategies for the various initial states. We remind the readers that we have considered $c_2 = \frac{\pi}{4}$, and hence, the two-qubit operators assume the form $J\left(\frac{\pi}{2} \geq c_1 \geq \frac{\pi}{4}, c_2 = \frac{\pi}{4}, c_3 = 0\right)$ in all the analysis. From the all the plots, it is clear that the average payoff of Alice remains unaffected when there is no noise in the channel ($p = 0$). Observe from Fig. 4a, d that for a given value of c_1 (for a particular entangling operator), the average payoff decreases with the increase in decoherence. Interestingly, for the initial states $|10\rangle$ and $|01\rangle$, when the strategies are $I \otimes \sigma_x$ and $\sigma_x \otimes I$, respectively, an increase in decoherence can increase the average payoff for the appropriate entangling operators. We can realize this fact by observing Fig. 4b, c

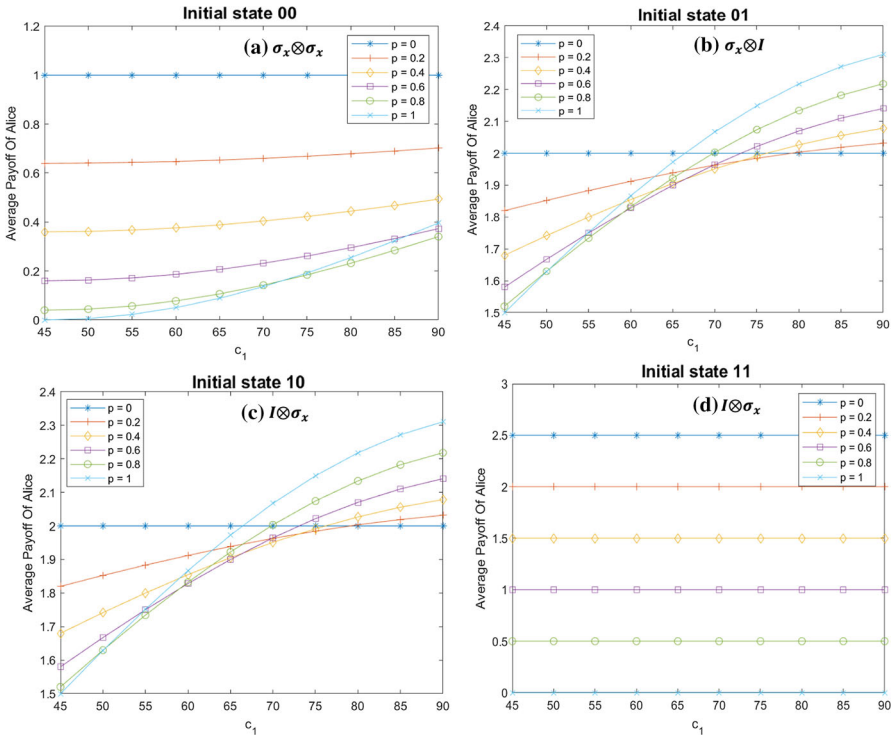

Fig. 4 Average payoff of Alice for different initial states and strategies when amplitude damping is present in channel 2

closely that increase in the parametric value c_1 of entangling operator increases the average payoff for a given value of p. Therefore, Fig. 4b, c suggests that payoff can be increased by tuning the entangling parameter even when the decoherence is maximum.

Note from this observation that entangling operator parametrized by c_1 either increases or decreases the average payoff for the given value of p. Thus, entangling operator J employed by the arbiter of the game cannot completely control the average payoff of the players in the prisoner's dilemma game. Such an observation has been arrived for the quantum two-penny flip game [14]. From the analysis carried out in 3.1 and 3.2, we can say in a qualitative way that decoherence in channel 2 affects the average payoff of the players more than when decoherence is in channel 1. A similar observation is found in the analysis of the prisoner's dilemma game which is studied using EML scheme [16].

4 Modified EWL scheme with decoherence and memory

Memory is the past successful knowledge of the player that is used to optimize their respective strategies. For the detailed study of memory in quantum games refer to [21–23]. When memory is present along with decoherence, one can use Kraus operators for the correlated noise and uncorrelated noise as [23]:

$$A_{00}^{c} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \sqrt{1-p} \end{pmatrix} \tag{18}$$

$$A_{00}^{u} = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-p} \end{pmatrix} \tag{19}$$

where $0 \leq p \leq 1$ is the measure of noise in the channel. The action of decoherence with memory for Alice and without memory for Bob are denoted as C_A and C_B, respectively. The expression of C_A and C_B becomes:

$$C_A = (1-\mu)A_{00}^{u} + \mu A_{00}^{c} \tag{20}$$

$$C_B = A_{00}^{u} \tag{21}$$

where μ is the amount of memory given to a player and $0 \leq \mu \leq 1$. While $\mu = 0$ represents a memoryless channel, $\mu = 1$ indicates a channel with maximum memory.

Consider the two-player two-strategy game, where only one player (say, Alice) is favored with memory which is represented as μ. To understand the memory and decoherence in the modified EWL scheme refer to Fig. 5. The final state of the game for this case is represented as:

$$\left|\psi_f\right\rangle = J^{\dagger} C_A C_B (U_A \otimes U_B) C_A C_B J |\psi_{\text{in}}\rangle \tag{22}$$

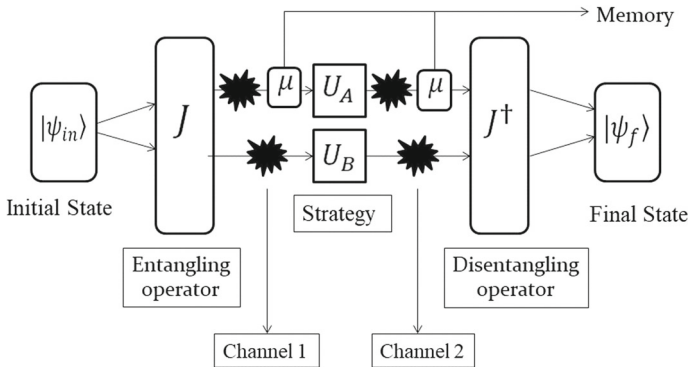

Fig. 5 Modified EWL scheme with memory and decoherence in channel 1 and channel 2

The quantum strategies of Alice and Bob, i.e., U_A and U_B from Eq. (2), considering Eqs. (20) and (21), $C_A C_B (U_A \otimes U_B) C_A C_B$ becomes:

$$C_A C_B (U_A \otimes U_B) C_A C_B = \begin{pmatrix} A & iaE & iaH^* & -bD^* \\ iaB & a^2F & -a^2G^* & iabC^* \\ iaC & -a^2G & a^2F^* & iabB^* \\ -bD & iabH & iabE^* & b^2A^* \end{pmatrix} \tag{23}$$

where $a = (1-\mu)(1-p) + \mu\sqrt{1-p}$ and $b = (1-\mu)(1-p)^2 + \mu(1-p)\sqrt{1-p}$.

The final state of the game for this case is obtained by substituting two-qubit entangling operator from Eq. (1) with $c_3 = 0$, initial state from Eqs. (3) and (23) in Eq. (22). After calculating the final state in thcomputational basis $|00\rangle$, $|01\rangle$, $|10\rangle$ and $|11\rangle$. we get:

$$\begin{aligned} \langle\psi_f|00\rangle = {} & a_1a_2\left(A^*c^-c^- + ibD^*s^-c^- - ibDc^-s^- + b^2As^-s^-\right) \\ & + a_1b_2\left(-iaE^*c^-c^+ - abH^*s^-c^+ + aHc^-s^+ - iabEs^-s^+\right) \\ & + b_1a_2\left(aE^*c^-s^+ - iabH^*s^-s^+ - iaHc^-c^+ - abEs^-c^+\right) \\ & + b_1b_2\left(iA^*c^-s^- - bD^*s^-s^- - bDc^-c^- - ib^2As^-c^-\right) \end{aligned} \tag{24}$$

$$\begin{aligned} \langle\psi_f|01\rangle = {} & a_1a_2\left(-iaB^*c^+c^- - aC^*s^+c^- + abCc^+s^- - iabBs^+s^-\right) \\ & + a_1b_2\left(a^2F^*c^+c^+ + ia^2G^*s^+c^+ - ia^2Gc^+s^+ + a^2Fs^+s^+\right) \\ & + b_1a_2\left(ia^2F^*c^+s^+ - a^2G^*s^+s^+ - a^2Gc^+c^+ - ia^2Fs^+c^+\right) \\ & + b_1b_2\left(aB^*c^+s^- - iaC^*s^+s^- - iabCc^+c^- - abBs^+c^-\right) \end{aligned} \tag{25}$$

$$\begin{aligned} \langle\psi_f|10\rangle = {} & a_1a_2\left(-aB^*s^+c^- - iaC^*c^+c^- - iabCs^+s^- + abBc^+s^-\right) \\ & + a_1b_2\left(-ia^2F^*s^+c^+ - a^2G^*c^+c^+ - a^2Gs^+s^+ + ia^2Fc^+s^+\right) \end{aligned}$$

$$+ b_1 a_2 \left(a^2 F^* s^+ s^+ - i a^2 G^* c^+ s^+ + i a^2 G s^+ c^+ + a^2 F c^+ c^+ \right)$$
$$+ b_1 b_2 \left(-i a B^* s^+ s^- + a C^* c^+ s^- - a b C s^+ c^- - i a b B c^+ c^- \right) \tag{26}$$

$$\begin{aligned}\langle \psi_f | 11 \rangle = {} & a_1 a_2 \left(-i A^* s^- c^- - b D^* c^- c^- - b D s^- s^- + i b^2 A c^- s^- \right) \\ & + a_1 b_2 \left(-a E^* s^- c^+ - i a b H^* c^- c^+ - i a H s^- s^+ + a b E c^- s^+ \right) \\ & + b_1 a_2 \left(-i a E^* s^- s^+ + a b H^* c^- s^+ - a H s^- c^+ - i a b E c^- c^+ \right) \\ & + b_1 b_2 \left(A^* s^- s^- - i b D^* c^- s^- + i b D s^- c^- + b^2 A c^- c^- \right)\end{aligned} \tag{27}$$

From the above expressions, the average payoff of Alice and Bob is calculated for the prisoner's dilemma game where X, Y, W and Z are the elements of the payoff matrix. The result is compiled in Table 3 for all the initial states and different strategies when memory is present in both the channels along with decoherence.

From Table 3, it is observed that for the initial state $|00\rangle$ and $|11\rangle$, the average payoff of the players is equal i.e., $\$_A = \$_B$ for the entangling operator $J\left(c_1 = \frac{\pi}{4}, c_2 = \frac{\pi}{4}, c_3 = 0\right)$ independent of the strategies adapted by the players. The identified entangling operator is a Double CNOT. Further, for the initial state $|01\rangle$ and $|10\rangle$, arbiter who applies the entangling operator plays no role in equalizing the average payoff of the players. This is similar to the case when decoherence is present in the channels. In this section, we try to analyse the role of memory when it is explicitly present in the channels along with decoherence graphically. From the observations of Table 3, the prisoner's dilemma game is analyzed using entangling operator (parameterized by c_1), decoherence (parametrized by p) and memory (parametrized by μ) through the plots shown in Fig. 6 for the initial state $|00\rangle$.

Once again, we emphasize that we have considered entangling operators $J\left(\frac{\pi}{2} \geq c_1 \geq \frac{\pi}{4}, c_2 = \frac{\pi}{4}, c_3 = 0\right)$ in all the analysis. Figure 6a, b is obtained by plotting the entangling operator (parametrized by c_1) and the average payoff of Alice for different values of p when $\mu = 0.2$ and $\mu = 0.8$, respectively, for the initial state $|00\rangle$ when the players adapt the strategy $I \otimes \sigma_x$. Observe from Fig. 6a, b that the average payoff of Alice decreases with the increase in decoherence for a particular choice of entangling operator. Further, the average payoff of Alice increases with the parametric value c_1 of entangling operator for the given value of p. Further, with the increase in memory, i.e., from $\mu = 0.2$ to $\mu = 0.8$, the effect of decoherence on the average payoff decreases to some extent for all values of p except for $p = 1$. Thus, memory cannot compensate for the effect of decoherence when it is maximum in the channel. Therefore, decrease in payoff due to decoherence is compensated to some extent by the increase in memory.

Figure 6c, d is obtained for the same initial state ($|00\rangle$) and for the same strategy ($I \otimes \sigma_x$) but for different values of μ when $p = 0.2$ and $p = 0.8$, respectively. It is evident from Fig. 6c, d that the increase in memory increases the average payoff of the player. In other words, as the degree of memory increases from 0 to 1, the effect of noise on the payoffs starts decreasing. However, an increase in the average payoff of the player is affected when the noise in the channel increases. Such an observation has also arrived from the work of Nawaz [14] when the entangling operator is CNOT. It is

Table 3 Average payoff of Alice and Bob when decoherence and memory are present in both the channels

Initial state	Strategies	Average payoff of Alice $\$_A$	Average payoff of Bob $\$_B$
$\|00\rangle$	$I \otimes I$	$X\left(c^-c^- + b^2s^-s^-\right)^2 + Z\left(c^-s^- - b^2c^-s^-\right)^2$	$X\left(c^-c^- + b^2s^-s^-\right)^2 + Z\left(c^-s^- - b^2c^-s^-\right)^2$
	$I \otimes \sigma_x$	$Ya^2\left(c^+c^- + bs^+s^-\right)^2 + Wa^2\left(c^-s^+ - bc^+s^-\right)^2$	$Wa^2\left(c^+c^- + bs^+s^-\right)^2 + Ya^2\left(c^-s^+ - bc^+s^-\right)^2$
	$\sigma_x \otimes I$	$Ya^2\left(s^+c^- - bc^+s^-\right)^2 + Wa^2\left(c^+c^- + bs^-s^+\right)^2$	$Wa^2\left(s^+c^- - bc^+s^-\right)^2 + Ya^2\left(c^+c^- + bs^-s^+\right)^2$
	$\sigma_x \otimes \sigma_x$	$Zb^2\left(c^-c^- + s^-s^-\right)^2$	$Zb^2\left(c^-c^- + s^-s^-\right)^2$
$\|11\rangle$	$I \otimes I$	$X\left(c^-s^- - b^2s^-c^-\right)^2 + Z\left(s^-s^- + b^2c^-c^-\right)^2$	$X\left(c^-s^- - b^2s^-c^-\right)^2 + Z\left(s^-s^- + b^2c^-c^-\right)^2$
	$I \otimes \sigma_x$	$Ya^2\left(c^+s^- - bs^+c^-\right)^2 + Wa^2\left(s^-s^+ + bc^+c^-\right)^2$	$Wa^2\left(c^+s^- - bs^+c^-\right)^2 + Ya^2\left(s^-s^+ + bc^+c^-\right)^2$
	$\sigma_x \otimes I$	$Ya^2\left(s^+s^- + bc^+c^-\right)^2 + Wa^2\left(c^+s^- - bc^-s^+\right)^2$	$Wa^2\left(s^+s^- + bc^+c^-\right)^2 + Ya^2\left(c^+s^- - bc^-s^+\right)^2$
	$\sigma_x \otimes \sigma_x$	$Xb^2\left(s^-s^- + c^-c^-\right)^2$	$Xb^2\left(s^-s^- + c^-c^-\right)^2$
$\|01\rangle$	$I \otimes I$	Ya^4	Wa^4
	$I \otimes \sigma_x$	$Xa^2\left(c^-c^+ + bs^-s^+\right)^2 + Za^2\left(c^+s^- - bc^-s^+\right)^2$	$Xa^2\left(c^-c^+ + bs^-s^+\right)^2 + Za^2\left(c^+s^- - bc^-s^+\right)^2$
	$\sigma_x \otimes I$	$Xa^2\left(s^+c^- - bc^+s^-\right)^2 + Za^2\left(s^-s^+ + bc^+c^-\right)^2$	$Xa^2\left(s^+c^- - bc^+s^-\right)^2 + Za^2\left(s^-s^+ + bc^+c^-\right)^2$
	$\sigma_x \otimes \sigma_x$	Wa^4	Ya^4
$\|10\rangle$	$I \otimes I$	Wa^4	Ya^4
	$I \otimes \sigma_x$	$Xa^2\left(c^-s^+ - bs^-c^+\right)^2 + Za^2\left(s^-s^+ + bc^+c^-\right)^2$	$Xa^2\left(c^-s^+ - bs^-c^+\right)^2 + Za^2\left(s^-s^+ + bc^+c^-\right)^2$
	$\sigma_x \otimes I$	$Xa^2\left(c^-c^+ + bs^-s^+\right)^2 + Za^2\left(c^+s^- - bc^-s^+\right)^2$	$Xa^2\left(c^-c^+ + bs^-s^+\right)^2 + Za^2\left(c^+s^- - bc^-s^+\right)^2$
	$\sigma_x \otimes \sigma_x$	Ya^4	Wa^4

also observed from the plots that for a given value of μ, the average payoff increases with an increase in parametric value c_1 of the entangling operator for all the values of decoherence.

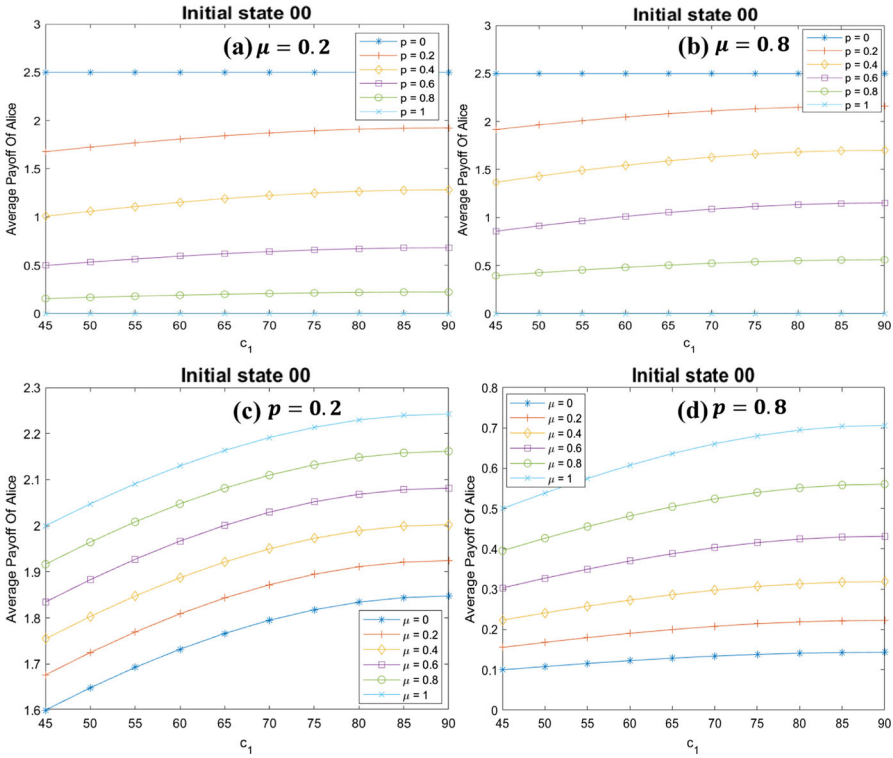

Fig. 6 Average payoff of Alice for the initial state $|00\rangle$ for the strategy $I \otimes \sigma_x$ when decoherence and memory are present in both channel 1 and channel 2

5 Conclusion

The dilemma in the classical prisoner's dilemma game is resolved under a very restricted case with the help of the quantization scheme. The well-known EWL scheme exploits only controlled unitary operators in the quantization procedure. This limitation has emerged due to the requirement of having a classical game as a subset of a quantum game. To understand the significance of other entangling operators, the modified EWL scheme is introduced by lifting the requirement in the EWL scheme [24]. When the quantum games are studied using EWL scheme, it is found that both decoherence and memory influences the average payoff of the players [20–23]. It is shown that decoherence decreases the average payoff of the players and only memory compensates for the decreases in payoff due to decoherence for the controlled-NOT operation [20–23]. In this work, both decoherence and memory are introduced in the modified EWL scheme as the scheme naturally admits many entangling operators, one can understand the interplay of entangling operators with memory and decoherence in the game structure.

As the result shown in [16], present work on modified EWL scheme indicates that the average payoff of players decreases with an increase in decoherence. However, the

present work reveals the fact that a decrease in average payoff due to decoherence can be minimized by the proper choice of entangling operator. Further, the presence of decoherence in channel 2 totally affects the average payoff of the players irrespective of the initial states and the strategies. On the other hand, when decoherence is in channel 1, its effect on the average payoff of the players is seen only for certain initial states and for certain strategies. Thus, decoherence in channel 2 affects the average payoff more when compared to decoherence in channel 1. This is due to the fact that decoherence in channel 2 occurs after the players have selected their strategies, whereas decoherence in channel 1 acts on the state before the players have chosen their strategies. This observation is consistent with the work [16]. However, it is clear from the present work that suitable use of the entangling operator can minimize the decrease in payoff due to decoherence in the channels. Further, increase in decoherence can increase the average payoff for the certain initial states and streategies when appropriate entangling operators are used.

Present work also establishes the fact that an increase in memory can increase the average payoff of the players depending upon the choice of entangling operator. When both memory and decoherence are present in the communication channels, memory controls the effect of decoherence [20–23]. Here we find that decrease in payoff due to decoherence can be compensated to some extent by increasing the memory. Once again, the availability of suitable entangling operators is shown to increase the payoff when decoherence and memory are present in the channels.

In a nutshell, we highlight three major observations from this work. First of all, the arbiter how has the entangling power cannot completely control the game under decoherence and memory irrespective of the initial states and strategies. This once again ensures the result presented in [14]. Secondly, decoherence plays a dominant role in the EWL scheme, and it is difficult to minimize its effect due to the consideration of only CNOT operation [16]. Whereas in the modified EWL scheme, it is possible to minimize the effect of decoherence by choosing appropriate entangling operators. Thirdly, memory always increases the average payoff of the players in the EWL scheme, and the effect of decoherence decreases as the memory increases [20–23]. Whereas in the modified EWL scheme, the effect of memory on the average payoff of the players can be tuned by choosing an appropriate entangling operator even though decoherence is present in the channels.

We have brought out a qualitative interplay among the entangling operators, decoherence and memory under the restricted situations of the game. To be precise, appropriate entangling operators are available to increase (decrease) the effect of memory (decoherence) on the average payoff of the players. Finding an exact relation among these parameters requires a regress mathematical treatment. Further, we have discussed only the amplitude damping channel in the modified EWL scheme as a starting analysis. The effect of decoherence using other noisy quantum communication channels, namely depolarising channel, phase damping channel, bit-flip channel warrant a detailed analysis. The modified EWL scheme which is implemented theoretically in this work, can be implemented in the form of a quantum circuit using IBM quantum experience, which is our future endeavor.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s11128-021-03216-8.

Acknowledgements Authors gratefully acknowledge the reviewers for their valuable comments and suggestions which improved the paper to the present form.

References

1. Myerson, R.B.: Game Theory: Analysis of Conflict. Harvard University Press (1992)
2. Von Neumann, J., Morgenstern, O.: Theory of Games and Economic Behavior. Princeton University Press (1947)
3. Guo, H., Zhang, J., Koehler, G.J.: A survey of quantum games. Decis. Support Syst. **46**(1), 318–332 (2008)
4. Meyer, D.A.: Quantum strategies. Phys. Rev. Lett. **82**, 1052–1055 (1999)
5. Eisert, J., Wilkens, M., Lewenstein, M.: Quantum games and quantum strategies. Phys. Rev. Lett. **83**, 3077–3080 (1999)
6. Benjamin, S.C., Hayden, P.M.: Multiplayer quantum games. Phys. Rev. A **64**, 030301(R) (2001)
7. Du, J., Xu, X., Li, H., Zhou, X., Han, R.: Entanglement playing a dominating role in quantum games. Phys. Letts. A. **289**, 9–15 (2001)
8. Elgazzar, A.S.: Quantum prisoner's dilemma in a restricted one parametric strategic space. Appl. Math. Comput. **370**, 124927 (2020)
9. Elgazzar, A.S.: Coopetition in quantum prisoner's dilemma and COVID-19. Quant. Info. Process. **20**, 102 (2021)
10. Flitney, A.P., Abbott, D.: Advantage of a quantum player over a classical one in 2×2 quantum games. Proc. R. Soc. Lond. A **459**, 2463–2474 (2003)
11. Schmid, C., Flitney, A.P., Wieczorek, W., Kiesel, N., Weinfurter, H., Hollenberg, L.C.L.: Experimental implementation of four player quantum game. New J. Phys. **12**, 063031 (2010)
12. Marinatto, L., Weber, T.: A quantum approach to static games of complete information. Phys. Lett. A **272**, 291–303 (2000)
13. Li, Q., Iqbal, A., Perc, M., Chen, M., Abbott, D.: Coevolution of quantum and classical strategies on evolving random networks. PLoS ONE **8**(7), e68423 (2013)
14. Sankrith, S., Dave, B., Balakrishnan, S.: Significance of entangling operators in quantum two penny flip game. Braz. J. Phys. **49**(6), 859–863 (2019)
15. Eisert, J., Wilkens, M.: Quantum games. J. Mod. Opt. **47**, 2543–2556 (2000)
16. Chen, L.K., Ang, H., Kiang, D., Kwek, L.C., Lo, C.F.: Quantum prisoner dilemma under decoherence. Phys. Letts. A **316**, 317–323 (2003)
17. Flitney, A.P., Abbott, D.: Quantum games with decoherence. J. Phys. A: Math. Gen. **38**, 449 (2005)
18. Khan, S., Ramzan, M., Khan, M.K.: Quantum Parrondo's game under decoherence. Int. J Theor. Phys **49**, 31 (2010)
19. Flitney, A.P., Hollenberg, L.C.L.: Multiplayer quantum minority game with decoherence. Quant. Inf. Comput. **7**(1–2), 111–127 (2007)
20. Nawaz, A.: The generalized quantization schemes for games and its application to quantum information, Ph.D. thesis, Quaid-I-Azam University, Islamabad, Pakistan (2007). http://arxiv.org/abs/quant-ph/1012.1933
21. Ramzan, M., Nawaz, A., Toor, A.H., Khan, M.K.: The effect of quantum memory on quantum games. J. Phys. A. Math. Theor. **41**(5), 055307 (2008)
22. Nawaz, A., Toor, A.H.: Quantum games with correlated noise. J. Phys. A: Math. Gen **39**, 9321 (2006)
23. Macchiavello, C., Palma, G.M.: Entanglement-enhanced information transmission over a quantum channel with correlated noise. Phys. Rev. A **65**, 050301 (2002)
24. Vijayakrishnan, V., Balakrishnan, S.: Role of two-qubit entangling operators in the modified Eisert–Wilkens–Lewenstein approach of quantization. Quant. Info. Process. **18**, 112 (2019)
25. Khan, F.S., Solmeyer, N., Balu, R., et al.: Quantum games: a review of the history, current state, and interpretation. Quant. Info. Process. **17**, 309 (2018)
26. Zhang, J., Vala, J., Whaley, K.B., Sastry, S.: Geometric theory of nonlocal two-qubit operations. Phys. Rev. A **67**, 042313 (2003)

27. Rezakhani, A.T.: Characterization of two-qubit perfect entanglers. Phys. Rev. A **70**, 052313 (2004)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.